Notes on Algebra

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Preface

These notes are motivated by the Algebra course taken by the author at the Institute of Mathematical Sciences (IMSc), Chennai, taught by Prof. Sankaran Vishwanath.

The purpose of these notes is to organise and develop the material covered in the course in a form suitable for future reference and study. These notes are not suitable for people studying group theory or rings and modules for the first time, since many arguments might be more hand-wavy for a first-time reader. However, the author believes these arguments contain enough information that people can complete them when required.

The preparation of this document involved substantial use of artificial intelligence tools for writing and exposition. However, the author has reviewed and verified everything to the best of their ability. Despite this effort, errors or inaccuracies may remain, and the notes should not be regarded as error-free.

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1 Monoids

1.1 Definition and Elementary Properties

Definition 1.1 (Monoid)**.** A **monoid** is a pair $(M, \cdot)$ consisting of a set M and a binary operation $\cdot : M \times M \to M$ satisfying the following two axioms:

1. **Associativity:** For all $a, b, c \in M$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. **Identity Element:** There exists an element $1_M \in M$ such that for all $m \in M$, $1_M \cdot m = m \cdot 1_M = m$.

1.2 Examples of Monoids and Homomorphisms

(a) The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ forms a monoid under multiplication, where the identity element is 1.

(b) The set $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ forms a monoid under addition, where the identity element is 0.

(c) Any group $(G, *)$ is naturally a monoid, as it satisfies associativity and possesses an identity element.

(d) If R is a ring with identity, then R forms a monoid under its multiplication operation. Specifically, the set of $n \times n$ matrices over a ring with identity, denoted $M_n(R)$, forms a monoid under matrix multiplication, where the identity element is the identity matrix I_n.

(e) The set of functions from a set X to X, denoted $\mathrm{End}(X) = X^X$, forms a monoid under function composition, where the identity map id_X serves as the identity element.

Definition 1.2 (Monoid Homomorphism)**.** Let M and N be monoids. A map $\varphi : M \to N$ is a **monoid homomorphism** if:

1. $\varphi(m_1 m_2) = \varphi(m_1)\varphi(m_2)$ for all $m_1, m_2 \in M$,
2. $\varphi(1_M) = 1_N$.

If a monoid homomorphism φ is bijective, it is called a **monoid isomorphism**. Two monoids M and N are said to be **isomorphic** (denoted $M \cong N$) if there exists a monoid isomorphism between them.

Exercise 1.1. Let $\mathcal{M}$ be the class of all monoids. Define a relation $\sim$ on $\mathcal{M}$ such that $M \sim N$ if and only if there exists a monoid isomorphism from M to N.

(a) Explain why $\mathcal{M}$ is a class and clearly not a set (such a class is called a **proper class**).
Hint: Russell's Paradox.

(b) Prove that $\sim$ is an equivalence relation on $\mathcal{M}$.

Exercise 1.2. Prove that the monoids $(\mathbb{N}, \cdot)$ and $(\mathbb{N}_0, +)$ given above are not isomorphic.

Exercise 1.3. Prove that the product of n monoid elements $a_1, a_2, \ldots, a_n$ taken in that exact order is independent of how the brackets are placed.
Hint: Use strong induction on n.

1.3 The Free Monoid and Universal Property

Definition 1.3 (Free Monoid)**.** Let X be an arbitrary set (often called an *alphabet*). The **free monoid** on X, denoted by X^*, is the set of all finite words (strings) formed by elements of X:

$$X^* = \{x_1 x_2 \cdots x_n \mid n \geq 0,\, x_i \in X\}.$$

When $n = 0$, the word is the empty word, denoted by 1 (or e).

Equivalently, we can construct X^* in terms of Cartesian products. Let $X_0 = \{1\}$, where 1 denotes the empty word of length 0. For each $n \geq 1$, let X_n denote the n-fold Cartesian product:

$$X_n = \underbrace{X \times X \times \cdots \times X}_{n \text{ times}} = \{(x_1, x_2, \ldots, x_n) \mid x_i \in X\}.$$

Identifying the ordered n-tuple $(x_1, x_2, \ldots, x_n) \in X_n$ with the concatenated word $x_1 x_2 \cdots x_n$, the free monoid is the disjoint union:

$$X^* = \coprod_{n=0}^{\infty} X_n.$$

The binary operation on X^* is string concatenation:

$$(x_1 \cdots x_n) \cdot (y_1 \cdots y_m) = x_1 \cdots x_n y_1 \cdots y_m,$$

which is associative, with the empty word 1 serving as the two-sided identity.

Example 1.1 (Free Monoid on One Generator)**.** Let $X = \{x\}$ be a singleton set. Every word in X^* consists solely of repeated concatenations of the single letter x:

$$X^* = \{1, x, x^2, x^3, \ldots\} = \{x^n \mid n \geq 0\},$$

where $x^0 = 1$ denotes the empty word. The concatenation operation obeys:

$$x^m \cdot x^n = x^{m+n} \quad \text{for all } m, n \geq 0.$$

In particular, $\{x\}^*$ is commutative. The map

$$\psi : \{x\}^* \longrightarrow (\mathbb{N}_0, +), \quad \psi(x^n) = n$$

is a bijection satisfying $\psi(x^m \cdot x^n) = m + n = \psi(x^m) + \psi(x^n)$ and $\psi(1) = 0$. Hence, the free monoid on one generator is isomorphic to the additive monoid of non-negative integers:

$$\{x\}^* \cong (\mathbb{N}_0, +).$$

Example 1.2 (Free Monoid on Two Generators)**.** Let $X = \{a, b\}$. The elements of X^* are all finite strings formed from the two-letter alphabet $\{a, b\}$. Grouped systematically by word length n:

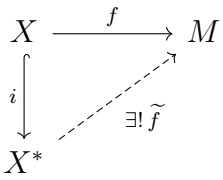
- **Length 0:** 1 (the empty word, $2^0 = 1$ element)
- **Length 1:** a, b ($2^1 = 2$ elements)
- **Length 2:** a^2, ab, ba, b^2 ($2^2 = 4$ elements)
- **Length 3:** $a^3, a^2b, aba, ab^2, ba^2, bab, b^2a, b^3$ ($2^3 = 8$ elements)

In general, there are 2^n distinct words of length n, growing exponentially.

Theorem 1.1 (Universal Property of Free Monoids)**.** *Let X be an arbitrary set, and let $i : X \hookrightarrow X^*$ be the canonical inclusion map sending $x \mapsto x$ (viewed as a word of length 1).*

For every monoid $(M, \cdot)$ and any set map $f : X \to M$, there exists a unique monoid homomorphism $\widetilde{f} : X^* \to M$ such that $\widetilde{f} \circ i = f$.

That is, the following diagram commutes:



Proof. **Existence:** Let $w \in X^*$. If $w = 1$ (the empty word), we define $\widetilde{f}(1) = 1_M$. If w is a non-empty word, it can be expressed uniquely as a finite sequence of letters:

$$w = x_1 x_2 \cdots x_k, \quad \text{where } k \geq 1 \text{ and } x_j \in X.$$

We define:

$$\widetilde{f}(w) = f(x_1) \cdot f(x_2) \cdots f(x_k) \in M.$$

To verify that $\widetilde{f}$ is a monoid homomorphism:

(i) *Operation preservation:* Let $u = x_1 \cdots x_k$ and $v = y_1 \cdots y_m$ be two non-empty words in X^*. Then $u \cdot v = x_1 \cdots x_k y_1 \cdots y_m$, and by associativity in M:

$$\widetilde{f}(u \cdot v) = f(x_1) \cdots f(x_k) f(y_1) \cdots f(y_m) = \widetilde{f}(u) \cdot \widetilde{f}(v).$$

If either $u = 1$ or $v = 1$, this identity holds trivially.

(ii) *Identity preservation:* By construction, $\widetilde{f}(1) = 1_M$.

Furthermore, for any letter $x \in X$, $i(x) = x$ is a word of length 1, so:

$$(\widetilde{f} \circ i)(x) = \widetilde{f}(x) = f(x).$$

Thus $\widetilde{f} \circ i = f$.

Uniqueness: Suppose $g : X^* \to M$ is another monoid homomorphism satisfying $g \circ i = f$. Because g is a homomorphism, $g(1) = 1_M = \widetilde{f}(1)$. For any word $w = x_1 x_2 \cdots x_k \in X^*$:

$$g(w) = g(x_1 x_2 \cdots x_k) = g(x_1)g(x_2) \cdots g(x_k).$$

Since $g \circ i = f$, we have $g(x_j) = g(i(x_j)) = f(x_j)$ for all $1 \leq j \leq k$. Hence:

$$g(w) = f(x_1)f(x_2) \cdots f(x_k) = \widetilde{f}(w).$$

Therefore, $g = \widetilde{f}$, establishing uniqueness. □

Proposition 1.1 (Properties of the Free Monoid)**.** *Let X be an arbitrary set and X^* the free monoid on X.*

(a) ***Cancellation Laws:*** *The free monoid X^* satisfies both left and right cancellation: for all words $u, v, w \in X^*$,*

$$wu = wv \implies u = v \quad \text{and} \quad uw = vw \implies u = v.$$

(b) ***Trivial Units:*** *The group of units (invertible elements) of X^* is trivial:*

$$(X^*)^\times = \{e\}.$$

In fact, if $u, v \in X^*$ satisfy $u \cdot v = e$, then $u = v = e$.

(c) ***Characterization of Commuting Words:*** *Two words $u, v \in X^*$ commute ($uv = vu$) if and only if they are integer powers of a common word:*

$$uv = vu \iff \exists\, w \in X^* \text{ and } k, m \geq 0 \text{ such that } u = w^k \text{ and } v = w^m.$$

Consequently, X^* is commutative if and only if $|X| \leq 1$. If $|X| \geq 2$, choosing distinct letters $a \neq b \in X$ yields $ab \neq ba$, so X^* is non-commutative.

Proof. Let $\ell : X \to (\mathbb{N}_0, +)$ denote the word length function defined by $\ell(x) = 1$ for $x \in X$. This induces a monoid homomorphism $\ell : X^* \to (\mathbb{N}_0, +)$ satisfying $\ell(u \cdot v) = \ell(u) + \ell(v)$ and $\ell(e) = 0$. Note that $\ell(x) = 0 \iff x = e$.

(a) *Cancellation:* Write $w = w_1 \cdots w_k$, $u = u_1 \cdots u_r$, and $v = v_1 \cdots v_s$, where all components are in X. The equality $wu = wv$ means:

$$w_1 \cdots w_k u_1 \cdots u_r = w_1 \cdots w_k v_1 \cdots v_s.$$

Comparing word lengths yields $k + r = k + s \implies r = s$. Comparing letters position by position at indices $k + 1$ through $k + r$ gives $u_j = v_j$ for all $1 \leq j \leq r$. Thus $u = v$. Right cancellation follows identically.

(b) *Units:* If $u \cdot v = e$, then $\ell(u \cdot v) = \ell(e) = 0$, so $\ell(u) + \ell(v) = 0$. Since $\ell(u), \ell(v) \in \mathbb{N}_0$, this forces $\ell(u) = \ell(v) = 0$, which implies $u = v = e$.

(c) *Commutativity:* We proceed by strong induction on $\max\{\ell(u), \ell(v)\}$. If $\ell(u) = 0$ or $\ell(v) = 0$, it follows trivially. Assume $\ell(u), \ell(v) \geq 1$ and $uv = vu$. Using the same argument as in (a), $uv = vu$, we must have $u = vt = tv$ or $v = ut = tu$ for some $t \in X^*$. Without loss of generality, assume $u = vt = tv$, then $\ell(t) = \ell(u) - \ell(v) < \ell(u)$. If $t = e$, we are done. Otherwise, by the induction hypothesis applied to $\{v, t\}$, there exists $w \in X^*$ such that $v = w^s$ and $t = w^p$. Then $u = vt = w^{s+p}$.

□

Example 1.3 (Failure of Free Monoid Properties in General Monoids)**.** The properties in Proposition 1.1 highlight the rigid nature of free monoids. General monoids can violate any of these conditions:

(a) **Failure of Cancellation:** In $(\mathbb{Z}, \cdot)$, cancellation fails due to 0:

$$0 \cdot 3 = 0 \cdot 5 = 0, \quad \text{yet } 3 \neq 5.$$

(b) **Failure of Trivial Units:** The group of invertible elements in $(\mathbb{Z}, \cdot)$ is:

$$(\mathbb{Z}, \cdot)^\times = \{1, -1\}.$$

(c) **Failure of the Common-Power Commutativity Characterization:** The monoid $(\mathbb{Z}, \cdot)$ is commutative, so $3 \cdot 5 = 5 \cdot 3 = 15$. However, 3 and 5 are distinct prime numbers; there is no base $w \in \mathbb{Z}$ and powers $k, m \geq 0$ such that $3 = w^k$ and $5 = w^m$.

Exercise 1.4. Find a monoid that is not a free monoid but satisfies properties (a), (b), and (c) of Proposition 1.1. Find particularly a non-abelian monoid.

Exercise 1.5 (Uniqueness of Free Monoids via the Universal Property)**.** Suppose M is a monoid containing X as a subset such that the triple (M, X, i) also satisfies the universal property of free monoids, where $i : X \hookrightarrow M$ is the inclusion map. That is, for every monoid N and every set map $f : X \to N$, there exists a unique monoid homomorphism $\widetilde{f} : M \to N$ such that $\widetilde{f} \circ i = f$.

Prove that M is isomorphic to X^* ($M \cong X^*$), and that this isomorphism preserves the generating set X pointwise.

1.4 Congruences and Quotient Monoids

Definition 1.4 (Multiplicative Equivalence Relation on Monoid)**.** Let $(M, \cdot)$ be a monoid. An equivalence relation $\sim$ on M is called a **multiplicative equivalence relation** (or a **monoid congruence**) if for all $w_1, w_2, w_1', w_2' \in M$:

$$w_1 \sim w_2 \quad \text{and} \quad w_1' \sim w_2' \implies w_1 w_1' \sim w_2 w_2'.$$

Equivalently, $\sim$ is a congruence if and only if it is compatible with multiplication on both sides:

$$a \sim b \implies ca \sim cb \quad \text{and} \quad ac \sim bc \quad \text{for all } a, b, c \in M.$$

Theorem 1.2 (Quotient Monoid)**.** *Let $(M, \cdot)$ be a monoid and let $\sim$ be an equivalence relation on M. Let $M/\sim = \{[x] \mid x \in M\}$ denote the set of equivalence classes under $\sim$.*

Define the binary operation on $M/\sim$ by:

$$[x] \cdot [y] := [xy] \quad \textit{for all } x, y \in M.$$

Then:

(a) The operation $[x] \cdot [y] = [xy]$ is well-defined if and only if $\sim$ is a congruence on M.

*(b) When $\sim$ is a congruence, $(M/\sim, \cdot)$ forms a monoid with identity element $[1_M]$, called the **quotient monoid**.*

Proof. **Part (a):**

- ($\implies$): Suppose $[x] \cdot [y] = [xy]$ is well-defined. If $w_1 \sim w_2$ and $w_1' \sim w_2'$, then $[w_1] = [w_2]$ and $[w_1'] = [w_2']$. Well-definedness guarantees $[w_1 w_1'] = [w_1] \cdot [w_1'] = [w_2] \cdot [w_2'] = [w_2 w_2']$, which means $w_1 w_1' \sim w_2 w_2'$. Hence $\sim$ is a congruence.
- ($\impliedby$): Suppose $\sim$ is a congruence. If $[x] = [x']$ and $[y] = [y']$, then $x \sim x'$ and $y \sim y'$, so by definition of a congruence, $xy \sim x'y'$. Thus $[xy] = [x'y']$, showing well-definedness.

Part (b):

(i) *Associativity:* $([x] \cdot [y]) \cdot [z] = [xy] \cdot [z] = [(xy)z] = [x(yz)] = [x] \cdot [yz] = [x] \cdot ([y] \cdot [z])$.

(ii) *Identity Element:* $[1_M] \cdot [x] = [1_M x] = [x]$ and $[x] \cdot [1_M] = [x 1_M] = [x]$.

Thus $(M/\sim, \cdot)$ is a monoid. □

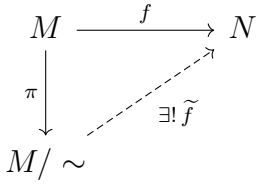
Theorem 1.3 (Universal Property of the Quotient Monoid)**.** *Let M be a monoid, $\sim$ a congruence on M, and $\pi : M \to M/\sim$ the canonical projection map defined by $\pi(x) = [x]$.*

The map π is a surjective monoid homomorphism. Furthermore, for every monoid N and every monoid homomorphism $f : M \to N$ satisfying:

$$x \sim y \implies f(x) = f(y) \quad \text{for all } x, y \in M,$$

*there exists a **unique** monoid homomorphism $\widetilde{f} : M/\sim \to N$ such that $\widetilde{f} \circ \pi = f$.*

That is, the following diagram commutes:



Proof. First, $\pi(xy) = [xy] = [x] \cdot [y] = \pi(x)\pi(y)$ and $\pi(1_M) = [1_M] = 1_{M/\sim}$, so π is a homomorphism. Surjectivity is immediate.

Existence: For $[x] \in M/\sim$, set $\widetilde{f}([x]) := f(x)$. If $[x] = [y]$, then $x \sim y$, so $f(x) = f(y)$; thus $\widetilde{f}$ is well-defined. Operation preservation and identity preservation follow directly from f:

$$\widetilde{f}([x][y]) = \widetilde{f}([xy]) = f(xy) = f(x)f(y) = \widetilde{f}([x])\widetilde{f}([y]),$$

and $\widetilde{f}([1_M]) = f(1_M) = 1_N$. Also $(\widetilde{f} \circ \pi)(x) = \widetilde{f}([x]) = f(x)$.

Uniqueness: If $g \circ \pi = f$, then $g([x]) = g(\pi(x)) = f(x) = \widetilde{f}([x])$, so $g = \widetilde{f}$. □

Exercise 1.6 (Intersections of Equivalence Relations and Congruences)**.** Let M be a monoid, and let $\mathcal{P}(M \times M)$ denote the power set of binary relations on M.

(a) Let $\{\sim_i\}_{i \in I}$ be an arbitrary non-empty family of equivalence relations on a set X, viewed as subsets $\sim_i \subseteq X \times X$. Prove that their intersection:

$$\sim := \bigcap_{i \in I} \sim_i$$

is also an equivalence relation on X.

(b) Suppose further that each $\sim_i$ is a **monoid congruence** on M (i.e., each $\sim_i$ is an equivalence relation compatible with the monoid operation on both sides). Prove that:

$$\sim := \bigcap_{i \in I} \sim_i$$

is also a monoid congruence on M.

(c) For any arbitrary relation $R \subseteq M \times M$, define:

$$\mathcal{C}(R) = \{\equiv \subseteq M \times M \mid \equiv \text{ is a monoid congruence on } M \text{ and } R \subseteq \equiv\}.$$

Show that $\mathcal{C}(R)$ is non-empty, and deduce that the **smallest congruence containing** R exists and is given explicitly by:

$$\sim_R := \bigcap_{\equiv \in \mathcal{C}(R)} \equiv .$$

Exercise 1.7 (Kernel Congruences and the First Isomorphism Theorem)**.** Let $f : M \to N$ be a monoid homomorphism.

(a) Define the relation $\sim_f$ on M by:

$$x \sim_f y \iff f(x) = f(y).$$

Prove that $\sim_f$ is a monoid congruence on M (called the **kernel congruence** of f, often denoted $\ker(f)$).

(b) Let $\pi : M \to M/\sim_f$ be the canonical projection homomorphism. Prove that there exists a unique injective monoid homomorphism $\bar{f} : M/\sim_f \to N$ such that $\bar{f} \circ \pi = f$. Deduce the **First Isomorphism Theorem for Monoids**:

$$M/\sim_f \cong \text{Im}(f).$$

(c) Let $\equiv$ be any monoid congruence on M. Prove that f factors through the canonical projection $\pi_\equiv : M \to M/\equiv$ (i.e., there exists a monoid homomorphism $\widetilde{f} : M/\equiv \to N$ such that $\widetilde{f} \circ \pi_\equiv = f$) if and only if:

$$\equiv \subseteq \sim_f .$$

In this sense, $\sim_f$ is the **largest (coarsest) congruence** on M through which f factors.

Exercise 1.8 (Congruences on Groups)**.** Let $(G, \cdot)$ be a group, and let $\sim$ be an equivalence relation on G that is a monoid congruence.

(a) Define $N := [1_G] = \{g \in G \mid g \sim 1_G\}$, the equivalence class of the identity element. Prove that N is a **normal subgroup** of G ($N \trianglelefteq G$).

(b) Prove that for all $x, y \in G$,

$$x \sim y \iff x^{-1}y \in N \iff xN = yN.$$

Thus, the equivalence classes of $\sim$ are precisely the cosets of N.

(c) Prove that the quotient monoid $G/\sim$ is a **group**, and is canonically isomorphic to the standard quotient group G/N.

(d) Conversely, prove that for any normal subgroup $N \trianglelefteq G$, the relation $\sim_N$ defined by $x \sim_N y \iff x^{-1}y \in N$ is a congruence on G.

Conclusion: The theory of monoid congruences on a group G coincides with the theory of normal subgroups and quotient groups. Every monoid congruence on a group is uniquely determined by a normal subgroup $N \trianglelefteq G$.

1.5 Divisibility, Chain Conditions, and Characterisation of Free Monoids

To characterize free monoids, we need to add conditions on monoids that ensure that every non-identity element can be factored into basic indecomposable elements (or atoms) uniquely.

In any monoid M with non-trivial invertible elements (units), the notion of irreducible factorization breaks down immediately: if $u \in M^\times$, one can artificially factor any element indefinitely as $x = (xu^{-1})u = (xu^{-2})u \cdot u$, and so on. To avoid this, we have the following approaches:

1. Define $u \sim v$ if $u = av$ or $u = va$ for some unit $a \in M^\times$. This is an equivalence relation (why?). We call u and v to be associates of each other. Then, define a non-unit element u as irreducible if whenever $u = xy$, either x or y is a unit.
2. Assume from the outset that the monoid has a trivial group of units: $M^\times = \{e\}$. In this case, associates collapse to equality ($u \sim v \iff u = v$).

To develop general theory, we will take the first approach even though free monoids have trivial group of units. Note that we will later see how our second condition will anyway give us a trivial group of units modulo a special case.

Definition 1.5 (Strict Divisibility and Irreducible Elements)**.** Let M be a monoid with identity element e, and let $M^\times$ denote its group of units.

(a) For $a, b \in M$, we say a **strictly divides** b (denoted $a \prec b$) if there exists $c \in M \setminus M^\times$ such that

$$b = ca \quad \text{or} \quad b = ac.$$

(b) An element $p \in M \setminus M^\times$ is called **irreducible** (or an **atom**) if whenever $p = ab$ for some $a, b \in M$, either $a \in M^\times$ or $b \in M^\times$. The set of all irreducible elements of M is denoted by $\mathcal{A}(M)$.

Exercise 1.9. Show that if u and v are associates of each other, then u is irreducible if and only if v is irreducible.

Even with trivial units, an element might still factor into non-units indefinitely without ever stopping at indecomposable components.

Example 1.4 (Failure of Divisor Termination in $(\mathbb{Q}_{\geq 0}, +)$)**.** Consider the additive monoid of non-negative rational numbers $(\mathbb{Q}_{\geq 0}, +)$ with identity element 0, which has trivial units $(\mathbb{Q}_{\geq 0})^\times = \{0\}$. Every element $q \in \mathbb{Q}_{>0}$ can be split indefinitely into non-zero sums:

$$q = \frac{q}{2} + \frac{q}{2} = \frac{q}{4} + \frac{q}{4} + \frac{q}{2} = \frac{q}{8} + \frac{q}{8} + \frac{q}{4} + \frac{q}{2} = \cdots$$

Because division by 2 can be iterated indefinitely, there exists an infinite strictly descending sequence of strict proper summands:

$$\cdots \prec \frac{q}{8} \prec \frac{q}{4} \prec \frac{q}{2} \prec q.$$

Consequently, $(\mathbb{Q}_{\geq 0}, +)$ contains no irreducible elements at all ($\mathcal{A}(\mathbb{Q}_{\geq 0}) = \emptyset$), and factorization into atomic elements is impossible.

To ensure that factorizations terminate, we impose a chain condition on divisors.

Definition 1.6 (Descending Chain Condition on Strict Divisors)**.** A monoid M satisfies the **descending chain condition (DCC) on strict divisors** if there is no infinite sequence of elements $(a_n)_{n \geq 1}$ such that each term strictly divides the preceding one:

$$\cdots \prec a_3 \prec a_2 \prec a_1.$$

That is, every strictly descending chain of divisors terminates in finitely many steps.

When the DCC on strict divisors holds alongside the condition $M \setminus M^\times \neq \emptyset$, the existence of atomic factorizations is guaranteed.

Proposition 1.2 (Existence of Factorization into Irreducibles)**.** *Let M be a monoid with $M \setminus M^\times \neq \emptyset$ satisfying the DCC on strict divisors. Then:*

(a) The set of irreducible elements $\mathcal{A}(M)$ is non-empty.

(b) Every element $x \in M \setminus M^\times$ can be expressed as a product of finitely many irreducible elements:

$$x = p_1 p_2 \cdots p_k, \quad \text{where } p_i \in \mathcal{A}(M) \text{ and } k \geq 1.$$

Proof. (a) Let $x \in M \setminus M^\times$. If $x \in \mathcal{A}(M)$, we are done. Otherwise, $x = a_1 b_1$ with $a_1, b_1 \in M \setminus M^\times$, so $a_1 \prec x$. If $a_1 \in \mathcal{A}(M)$, we stop; otherwise, we factor a_1 further to obtain a chain $\cdots \prec a_2 \prec a_1 \prec x$. By the DCC on strict divisors, this chain cannot be infinite and must terminate at some element $p \in M \setminus M^\times$ that admits no non-trivial factorization into non-units. Hence $p \in \mathcal{A}(M)$, proving $\mathcal{A}(M) \neq \emptyset$.

(b) Suppose there exists $x \in M \setminus M^\times$ that cannot be written as a finite product of irreducibles. Let $S \subseteq M \setminus M^\times$ be the set of all such elements. Since $x \in S$, $S \neq \emptyset$.

By the DCC on strict divisors, every non-empty subset of M contains an element that is minimal with respect to $\prec$ (otherwise, one could indefinitely pick strictly smaller divisors within S, generating an infinite descending chain $\cdots \prec s_2 \prec s_1$, violating DCC).

Let $m \in S$ be a minimal element of S. Since $m \in S$, m cannot be irreducible, so $m = uv$ for some non-units $u, v \in M \setminus M^\times$. Then $u \prec m$ and $v \prec m$. By the minimality of m in S, neither u nor v can belong to S. Therefore, both u and v can be factored into finite products of irreducibles:

$$u = p_1 \cdots p_r \quad \text{and} \quad v = q_1 \cdots q_s \quad (p_i, q_j \in \mathcal{A}(M)).$$

Thus $m = uv = p_1 \cdots p_r q_1 \cdots q_s$, expressing m as a finite product of irreducibles, which contradicts $m \in S$. Hence $S = \emptyset$, and every non-unit factors into irreducibles. □

While trivial units and the DCC on divisors guarantee that *some* atomic factorization exists, they do **not** guarantee that factorizations are unique.

Proposition 1.3 (Failure of Uniqueness: $ab = cd$)**.** *Let $X = \{a, b, c, d\}$ and consider the monoid M presented by generators and relations:*

$$M = \langle a, b, c, d \mid ab = cd \rangle \cong X^*/\sim,$$

where $\sim$ is the smallest monoid congruence on the free monoid X^* containing the pair (ab, cd). Then:

(a) M admits a well-defined length function $\ell : M \to (\mathbb{N}_0, +)$ satisfying $\ell(x) = 0 \iff x = e$. Consequently, M has trivial units ($M^\times = \{e\}$) and satisfies the DCC on strict divisors.

(b) The irreducible elements of M are precisely the images of the generators:

$$\mathcal{A}(M) = \{[a], [b], [c], [d]\}.$$

(c) The elements $[a], [b], [c], [d]$ are pairwise distinct in M. In particular, the element $x = [ab] = [cd]$ admits two distinct factorizations into irreducibles:

$$[a] \cdot [b] = [c] \cdot [d] \quad \text{with } [a] \neq [c] \text{ and } [b] \neq [d].$$

Consequently, factorization into irreducibles is not unique, and M cannot be isomorphic to any free monoid.

Proof. (a) Let $\ell_0 : X^* \to (\mathbb{N}_0, +)$ be the canonical word length homomorphism defined by $\ell_0(x) = 1$ for all $x \in X$. Consider the equivalence relation induced by ℓ_0:

$$R_\ell := \{(u, v) \in X^* \times X^* \mid \ell_0(u) = \ell_0(v)\}.$$

Since ℓ_0 is a monoid homomorphism, R_ℓ is a monoid congruence on X^*. Evaluating ℓ_0 on the pair (ab, cd):

$$\ell_0(ab) = 1 + 1 = 2 = 1 + 1 = \ell_0(cd) \implies (ab, cd) \in R_\ell.$$

Because $\sim$ is, by definition, the *smallest* congruence on X^* containing (ab, cd), and R_ℓ is a congruence containing (ab, cd), it follows by minimality that:

$$\sim \subseteq R_\ell.$$

Therefore, $u \sim v \implies \ell_0(u) = \ell_0(v)$. By the universal property of the quotient monoid, ℓ_0 factors uniquely through $M = X^*/\sim$ to induce a well-defined homomorphism $\ell : M \to (\mathbb{N}_0, +)$ given by $\ell([w]) = \ell_0(w)$. This also proves $\ell([x]) = 0 \iff [x] = e$.

(b) Note that $[x] \in \{[a], [b], [c], [d]\} \iff \ell([x]) = 1$. If $[x] = uv$ for $u, v \in M$, then $\ell(u) + \ell(v) = 1$. Since $\ell(u), \ell(v) \in \mathbb{N}_0$, either $\ell(u) = 0 \implies u = e$ or $\ell(v) = 0 \implies v = e$. Thus $[x]$ cannot factor into two non-identity elements, so $\{[a], [b], [c], [d]\} \subseteq \mathcal{A}(M)$.

Conversely, any element $w \in M$ with $\ell(w) \geq 2$ can be represented as $w = [x_1]([x_2] \cdots [x_k])$ with $k \geq 2$ and $x_1, \ldots, x_k \in X$. Each factor has length ≥ 1 (hence non-identity). Thus no element of length ≥ 2 is irreducible, giving $\mathcal{A}(M) = \{[a], [b], [c], [d]\}$.

(c) To prove $[a] \neq [c]$, define a set map $f : X \to (\mathbb{N}_0, +)$ by sending:

$$f(a) = 0, \quad f(d) = 0, \quad f(b) = 1, \quad f(c) = 1.$$

By the universal property of the free monoid, f extends uniquely to a homomorphism $\widetilde{f} : X^* \to (\mathbb{N}_0, +)$. Consider its kernel congruence:

$$R_f := \{(u, v) \in X^* \times X^* \mid \widetilde{f}(u) = \widetilde{f}(v)\}.$$

Evaluating $\widetilde{f}$ on the relation words:

$$\widetilde{f}(ab) = f(a) + f(b) = 0 + 1 = 1 = 1 + 0 = f(c) + f(d) = \widetilde{f}(cd).$$

Thus $(ab, cd) \in R_f$. Since $\sim$ is the smallest congruence on X^* containing (ab, cd), minimality forces:

$$\sim \subseteq R_f.$$

Therefore, $\widetilde{f}$ factors through M to induce a well-defined homomorphism $\overline{f} : M \to (\mathbb{N}_0, +)$ satisfying $\overline{f}([w]) = \widetilde{f}(w)$. Evaluating at $[a]$ and $[c]$:

$$\overline{f}([a]) = f(a) = 0 \quad \text{and} \quad \overline{f}([c]) = f(c) = 1.$$

Because $\overline{f}([a]) \neq \overline{f}([c])$, it follows that $[a] \neq [c]$ in M.

Thus, the element $x = [ab] = [cd]$ admits two distinct factorizations into irreducibles:

$$[a] \cdot [b] = [c] \cdot [d] \quad \text{with } [a] \neq [c] \text{ and } [b] \neq [d].$$

In any free monoid Y^*, every element has a strictly unique factorization into elements of its generating alphabet $Y = \mathcal{A}(Y^*)$. Because factorization into atoms fails to be unique in M, M cannot be isomorphic to any free monoid. $\square$

To eliminate this branching and ensure unique factorisation, we require that leading and trailing irreducible factors are rigidly determined.

Definition 1.7 (Irreducible Cancellation Conditions)**.** Let M be a monoid with group of units $M^\times$ and set of irreducible elements $\mathcal{A}(M)$.

(a) **Strong Prefix/Suffix Cancellation:**

- M satisfies **strong prefix-cancellation on irreducibles** if for all $a, b \in \mathcal{A}(M)$ and $w, w' \in M$:
$$aw = bw' \implies a = b \quad \text{and} \quad w = w'.$$
- Symmetrically, M satisfies **strong suffix-cancellation on irreducibles** if for all $a, b \in \mathcal{A}(M)$ and $w, w' \in M$:
$$wa = w'b \implies a = b \quad \text{and} \quad w = w'.$$

(b) **Weak Prefix/Suffix Condition (Uniqueness up to Associates):**

- M satisfies the **weak prefix condition on irreducibles** if for all $a, b \in \mathcal{A}(M)$ and $w, w' \in M$:
$$aw = bw' \implies a \sim b.$$
- Symmetrically, M satisfies the **weak suffix condition on irreducibles** if for all $a, b \in \mathcal{A}(M)$ and $w, w' \in M$:
$$wa = w'b \implies a \sim b.$$

Definition 1.8 (Levi's Condition / Equidivisibility)**.** A monoid M satisfies **Levi's condition** (or is called **equidivisible**) if for all $x, y, z, w \in M$,

$$xy = zw \implies \exists\, u \in M \text{ such that } (x = zu \text{ and } uw = y) \text{ or } (z = xu \text{ and } uy = w).$$

Exercises

Exercise 1.10 (Units and Strong Cancellation)**.** Let M be a monoid with $M \setminus M^\times \neq \emptyset$ satisfying the DCC on strict divisors.

(a) Suppose M satisfies strong prefix-cancellation on irreducibles. Prove that M must have a trivial group of units:
$$M^\times = \{e\}.$$

(b) Explain why the statement in (a) fails if M is a non-trivial group, and identify what $\mathcal{A}(M)$ is in that case.

Exercise 1.11 (Levi's Condition and Weak Cancellation on Irreducibles)**.** Let M be an arbitrary monoid (with possibly non-trivial units $M^\times$).

(a) Prove that if M satisfies Levi's condition, then M satisfies both the **weak prefix condition** and the **weak suffix condition** on irreducibles:
$$\text{For } a, b \in \mathcal{A}(M), \quad aw = bw' \implies a \sim b \quad \text{and} \quad wa = w'b \implies a \sim b.$$

(b) Deduce that if M additionally has trivial units ($M^\times = \{e\}$), then:

$$a \sim b \iff a = b.$$

Conclude that in any monoid with trivial units, Levi's condition implies both strong prefix- and strong suffix-cancellation on irreducibles.

Exercise 1.12 (Uniqueness of Factorization and Length Homomorphism)**.** Let M be a monoid with $M \setminus M^\times \neq \emptyset$ satisfying the DCC on strict divisors and strong prefix-cancellation on irreducibles.

(a) Prove that every element $x \in M \setminus \{e\}$ has a **unique** expression as a product of irreducible elements: if

$$a_1 a_2 \cdots a_k = b_1 b_2 \cdots b_m \quad (a_i, b_j \in \mathcal{A}(M)),$$

then $k = m$ and $a_i = b_i$ for all $1 \leq i \leq k$.

(b) Deduce that the length map $\ell : M \to (\mathbb{N}_0, +)$ defined by $\ell(e) = 0$ and $\ell(p_1 \cdots p_k) = k$ is a well-defined monoid homomorphism satisfying $\ell(x) = 0 \iff x = e$.

Exercise 1.13 (Characterization Theorem for Free Monoids)**.** Let M be a monoid. Prove that the following conditions are equivalent:

(i) M is a free monoid with $M \cong X^*$ for some non-empty set X.

(ii) M satisfies the following three conditions:

- $M \neq M^\times$,
- M satisfies the descending chain condition on strict divisors,
- M satisfies strong prefix-cancellation (or strong suffix-cancellation) on irreducibles.

(iii) M satisfies the following three conditions:

- $M^\times = \{e\}$,
- M satisfies the descending chain condition on strict divisors,
- M satisfies weak suffix-cancellation on irreducibles.

(iv) $M^\times = \{e\}$, satisfies the DCC on strict divisors, and satisfies Levi's condition.

(v) M admits a length map $\ell : M \to (\mathbb{N}_0, +)$ with $\ell(x) = 0 \iff x = e$, and satisfies Levi's condition.

Furthermore, deduce that any such monoid satisfies:

(a) **Cancellation Laws:** $wu = wv \implies u = v$ and $uw = vw \implies u = v$.

(b) **Commuting Elements:** Two elements $u, v \in M$ commute ($uv = vu$) if and only if they are powers of a common element:

$$uv = vu \iff \exists\, w \in M,\ k, m \geq 0 \text{ such that } u = w^k \text{ and } v = w^m.$$

Exercise 1.14.

(a) **Monoids as Quotients of Free Monoids:** Prove that every monoid M is isomorphic to a quotient of a free monoid X^* by a congruence relation $\sim$.

(b) **Irreducibles and Factorization in Submonoids:** Let $M = X^*$ be a free monoid, and let $N \subseteq M$ be a non-trivial submonoid with the same identity element ($e \in N$). Let Σ_N denote the set of irreducible elements of N. Prove that:

(i) Σ_N is non-empty.

(ii) Every non-identity element $w \in N$ can be written as a finite product of irreducible elements in Σ_N.

(c) **Freeness of Submonoids:** By the universal property of free monoids, the inclusion map $\Sigma_N \hookrightarrow N$ extends uniquely to a surjective monoid homomorphism:

$$\Phi : \Sigma_N^* \longrightarrow N, \quad \Phi(s_1 \cdots s_k) = s_1 \cdots s_k.$$

(a) Is every submonoid of a free monoid free? That is, is Φ always injective? If not, provide a counterexample.

(b) When is Φ injective (i.e., when is N free)?
(Hint: Use Levi's Condition or strong cancellation)

(d) **Submonoids of $(\mathbb{N}_0, +)$:** Recall that $(\mathbb{N}_0, +) \cong \{x\}^*$ is the free monoid on one generator.

(a) Consider the submonoid $S = \{3n + 4m + 5k \mid n, m, k \in \mathbb{N}_0\} = \langle 3, 4, 5 \rangle \subseteq \mathbb{N}_0$. Determine its set of irreducible elements. Is S a free monoid?

(b) More generally, determine the precise necessary and sufficient condition for a submonoid $S \subseteq (\mathbb{N}_0, +)$ to be a free monoid.

(e) If we define a submonoid of a given monoid to be a subset that forms a monoid under restriction of the binary operation. Can it have a different identity? Formulate a necessary and sufficient condition in terms of the new identity. Does a free monoid element satisfy the condition? Can we drop the same identity element in (b)?

1.6 Construction and Universal Property of the Free Group

In Section 1, we constructed the free monoid X^* as the set of all finite words over an alphabet X under concatenation. To construct a group from X, we must adjoin formal inverses for every element of X and enforce the relation that adjacent inverse pairs reduce to the identity element.

Definition 1.9 (Formal Inverses and Free Group Construction)**.** Let X be an arbitrary set.

1. Let $X' = \{x' \mid x \in X\}$ be a disjoint copy of X, representing formal inverses (where x' serves as the symbol for x^{-1}). Define the extended alphabet:
$$\overline{X} := X \cup X'.$$
2. Define the involution map $'$ on $\overline{X}$ by setting $(x)' = x'$ and $(x')' = x$ for all $x \in X$.
3. Consider the free monoid $\overline{X}^*$ under word concatenation. We define congruence $\widehat{R}$ on $\overline{X}^*$ to be the smallest congruence that contains the relation R:
$$R = \{(xx', 1),\ (x'x, 1) \mid x \in X\}.$$
Thus, $\widehat{R}$ is the unique smallest monoid congruence on $\overline{X}^*$ identifying adjacent inverse symbols with the empty word 1.
4. The **free group on** X is defined as the quotient monoid:
$$FG(X) := \overline{X}^*/\widehat{R}.$$

Proposition 1.4 ($FG(X)$ is a Group)**.** *The quotient monoid $FG(X) = \overline{X}^*/\widehat{R}$ is a group.*

Proof. By Theorem 1.2, $FG(X)$ is already a monoid.

(i) **Existence of Inverses:** Let $[w] \in FG(X)$ with representative word $w \in \overline{X}^*$. If $w = 1$, then $[1][1] = [1]$. If w is a non-empty word, it can be written as:
$$w = u_1 u_2 \cdots u_n \quad \text{with each } u_j \in \overline{X}.$$
Define the reversed word with inverted letters:
$$w' := u_n' \cdots u_2' u_1' \in \overline{X}^*.$$
Concatenating w and w' in the free monoid $\overline{X}^*$ gives:
$$ww' = u_1 \cdots u_{n-1}(u_n u_n')u_{n-1}' \cdots u_1'.$$
By definition of $\widehat{R}$, $(u_n u_n', 1) \in \widehat{R}$. Because $\widehat{R}$ is a monoid congruence, it is compatible with concatenation on both sides so:
$$ww' \sim u_1 \cdots u_{n-1} u_{n-1}' \cdots u_1'.$$

Iterating this reduction inside-out n times yields:

$$ww' \sim u_1 \cdots (u_{n-1}u'_{n-1}) \cdots u'_1 \sim \cdots \sim u_1 u'_1 \sim 1,$$

i.e.,

$$[w] \cdot [w'] = [ww'] = [1].$$

Similarly, $[w'][w] = [w'w] = [1]$. Therefore, every element $[w] \in FG(X)$ possesses a two-sided inverse $[w]^{-1} = [w']$, completing the proof. □

Before proving the universal property, we examine the concrete structure of $FG(X)$ for alphabets of small cardinality.

Example 1.5 (Free Group on One Letter)**.** Let $X = \{x\}$ be a singleton alphabet. Then $X' = \{x'\}$, and the extended alphabet is $\overline{X} = \{x, x'\}$.

In the free monoid $\overline{X}^*$, every word is an arbitrary string of x's and x''s. Under the congruence $\widehat{R}$, we have the relations:

$$xx' \sim 1 \quad \text{and} \quad x'x \sim 1.$$

Because the letters commute up to inverse cancellation (x and x' eliminate each other whenever they are adjacent), every word $w \in \overline{X}^*$ reduces to a unique word of the form:

$$\begin{cases} x^n = \underbrace{xx \cdots x}_{n \text{ times}} & \text{if } n > 0, \\ 1 & \text{if } n = 0, \\ (x')^{|n|} = \underbrace{x'x' \cdots x'}_{|n| \text{ times}} & \text{if } n < 0. \end{cases}$$

Writing x^{-1} in place of x' and x^{-k} for $(x')^k$, every element of $FG(\{x\})$ can be written uniquely as x^n for some $n \in \mathbb{Z}$. The group operation satisfies:

$$[x^n] \cdot [x^m] = [x^{n+m}] \quad \text{for all } n, m \in \mathbb{Z}.$$

Consequently, the map $\psi : FG(\{x\}) \to (\mathbb{Z}, +)$ defined by $\psi([x^n]) = n$ is a group isomorphism:

$$FG(\{x\}) \cong (\mathbb{Z}, +).$$

In particular, the free group on a single letter is abelian and infinite cyclic.

Example 1.6 (Free Group on Two Letters)**.** Let $X = \{a, b\}$. Then $X' = \{a', b'\}$ (which we write as $\{a^{-1}, b^{-1}\}$), so the extended alphabet is $\overline{X} = \{a, b, a^{-1}, b^{-1}\}$.

Unlike the single-letter case, the generators a and b do not commute. The only allowed cancellations in $\overline{X}^*$ are adjacent pairs of the same letter:

$$aa^{-1} \sim 1, \quad a^{-1}a \sim 1, \quad bb^{-1} \sim 1, \quad b^{-1}b \sim 1.$$

A word in $\overline{X}^*$ is called **reduced** if it contains none of the subwords aa^{-1}, $a^{-1}a$, bb^{-1}, or $b^{-1}b$. Every equivalence class in $FG(\{a, b\})$ contains a unique reduced word:

- **Length 0:** 1 (1 element).

- **Length 1:** a, a^{-1}, b, b^{-1} (4 elements).
- **Length 2:** Words of length 2 where no letter is followed by its inverse:

$$a^2,\ ab,\ ab^{-1}, \quad a^{-2},\ a^{-1}b,\ a^{-1}b^{-1}, \quad b^2,\ ba,\ ba^{-1}, \quad b^{-2},\ b^{-1}a,\ b^{-1}a^{-1}$$

  $(4 \times 3 = 12$ elements).
- **Length** $n \geq 1$**:** Each subsequent letter has $2|X| - 1 = 3$ choices (any symbol except the inverse of the preceding letter). Hence, there are:

$$4 \cdot 3^{n-1}$$

  distinct reduced words of length n.

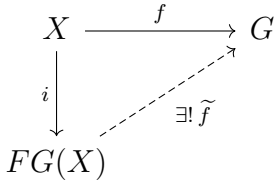
In particular, $ab \not\sim ba$ because neither ab nor ba contains any adjacent inverse pairs, so $[a][b] \neq [b][a]$. Thus, $FG(\{a, b\})$ is non-abelian and infinite.

Theorem 1.4 (Universal Property of Free Groups)**.** *Let X be an arbitrary set, and let $i : X \to FG(X)$ be the canonical map given by $i(x) = [x]$.*

For every group G and every set map $f : X \to G$, there exists a ***unique*** *group homomorphism $\widetilde{f} : FG(X) \to G$ such that:*

$$\widetilde{f} \circ i = f.$$

That is, the following diagram commutes:



Proof. **Existence:**

1. Define a map on the extended alphabet $\overline{f} : \overline{X} \to G$ by:

$$\overline{f}(x) = f(x) \quad \text{and} \quad \overline{f}(x') = (f(x))^{-1} \quad \text{for all } x \in X.$$

2. By the Universal Property of Free Monoids (Theorem 1.1), there exists a unique monoid homomorphism $f^* : \overline{X}^* \to G$ extending $\overline{f}$:

$$f^*(u_1u_2 \cdots u_n) = \overline{f}(u_1)\overline{f}(u_2) \cdots \overline{f}(u_n) \in G, \quad \text{and} \quad f^*(1) = 1_G.$$

   Evaluating f^* on xx' and $x'x$:

$$f^*(xx') = f^*(x)f^*(x') = f(x)(f(x))^{-1} = 1_G = f^*(1),$$

$$f^*(x'x) = f^*(x')f^*(x) = (f(x))^{-1}f(x) = 1_G = f^*(1).$$

3. By the Universal Property of Quotient Monoids (Theorem 1.3), there exists a unique monoid homomorphism $\widetilde{f} : FG(X) \to G$ such that $\widetilde{f}([w]) = f^*(w)$ for all $w \in \overline{X}^*$. Since $FG(X)$ and G are groups, any monoid homomorphism between groups is automatically a group homomorphism.

4. For any $x \in X$, $(\widetilde{f} \circ i)(x) = \widetilde{f}([x]) = f^*(x) = \overline{f}(x) = f(x)$.

Uniqueness: Suppose $g : FG(X) \to G$ is another group homomorphism satisfying $g \circ i = f$. For any $x \in X$, $g([x]) = g(i(x)) = f(x) = \widetilde{f}([x])$. Since g is a group homomorphism, it preserves inverses:

$$g([x']) = g([x]^{-1}) = (g([x]))^{-1} = (f(x))^{-1} = \widetilde{f}([x']).$$

Now,

$$g([u_1 \cdots u_n]) = g([u_1]) \cdots g([u_n]) = \widetilde{f}([u_1]) \cdots \widetilde{f}([u_n]) = \widetilde{f}([u_1 \cdots u_n]).$$

Hence $g = \widetilde{f}$, establishing uniqueness. □

Corollary 1.1 (Uniqueness of Free Groups up to Isomorphism)**.** *Let X be a set. If F is any group equipped with a set map $j : X \to F$ satisfying the universal property of free groups on X, then there exists a unique group isomorphism $\theta : FG(X) \xrightarrow{\sim} F$ such that $\theta \circ i = j$.*

Proof. By the universal property of $FG(X)$ applied to the map $j : X \to F$, there exists a unique group homomorphism $\theta : FG(X) \to F$ such that $\theta \circ i = j$. Conversely, by the universal property of F applied to $i : X \to FG(X)$, there exists a unique group homomorphism $\psi : F \to FG(X)$ such that $\psi \circ j = i$.

Composing them:

$$(\psi \circ \theta) \circ i = \psi \circ j = i = \mathrm{id}_{FG(X)} \circ i.$$

By the unique map in the universal property of $FG(X)$ applied to the set map $i : X \to FG(X)$, the only endomorphism of $FG(X)$ commuting with i is the identity map; hence $\psi \circ \theta = \mathrm{id}_{FG(X)}$. Similarly, $(\theta \circ \psi) \circ j = j = \mathrm{id}_F \circ j$ forces $\theta \circ \psi = \mathrm{id}_F$. Thus θ is a group isomorphism. □

1.7 Smallest Monoid Congruences and the Reduced Word Scheme

1.7.1 Top-Down vs. Bottom-Up View of $\widehat{R}$

For a monoid M, while defining $M/\widehat{R}$ (the smallest congruence containing R), the relation $\widehat{R}$ is obtained through two equivalent perspectives:

1. **Top-Down (Intersection):** $\widehat{R}$ is the intersection of all monoid congruences containing R. This guarantees that $\widehat{R}$ is minimal, making factorisations through quotient monoids automatic.

2. **Bottom-Up (Elementary Rewriting Chains):** Define the symmetric 1-step reduction relation $\longleftrightarrow$ on M by:

   $$u \longleftrightarrow v \iff \exists\, p, q \in M,\ (a, b) \in R \quad \text{such that} \quad u = paq \ \text{ and } \ v = pbq \ \text{ (or vice-versa).}$$

   Define $u \sim v$ if and only if there exists a finite chain ($k = 0$ allowed):

   $$u = w_0 \longleftrightarrow w_1 \longleftrightarrow w_2 \longleftrightarrow \cdots \longleftrightarrow w_k = v.$$

   Notice that $u \sim v \implies u\widehat{R}v$. Also, $\sim$ is a congruence on the monoid M. Hence, $\sim = \widehat{R}$.

For $FG(X)$,

$$u \longleftrightarrow v \iff \exists p, q \in \overline{X}^*, (a, a') \in X \quad \text{such that} \quad u = paa'q \ \text{ and } \ v = pq \ \text{ (or vice-versa)}.$$

1.7.2 Normal Forms: Reduced Words

Definition 1.10 (Reduced Word)**.** A word $w = u_1 u_2 \cdots u_n \in \overline{X}^*$ (where each $u_j \in \overline{X}$) is called **reduced** if it contains no adjacent formal inverse pairs, that is:

$$u_{j+1} \neq u_j' \quad \text{for all } 1 \leq j \leq n-1.$$

The empty word 1 (length 0) is reduced. Let $\mathcal{R}(X) \subseteq \overline{X}^*$ denote the set of all reduced words.

Theorem 1.5 (Normal Form Theorem for Free Groups)**.** *Every congruence class $[w] \in FG(X)$ contains **exactly one** reduced word. That is, the canonical projection map restricted to reduced words:*

$$\pi|_{\mathcal{R}(X)} : \mathcal{R}(X) \longrightarrow FG(X), \quad w \longmapsto [w]$$

is a bijection.

Proof. **Surjectivity (Existence):** Let $w \in \overline{X}^*$. If w is not reduced, it contains a subword yy' with $y \in \overline{X}$. Replacing yy' by 1 produces a strictly shorter word w_1 such that $w \ \widehat{R} \ w_1$ with $\ell(w_1) = \ell(w) - 2$. Because word lengths are non-negative integers, this reduction process terminates in a finite number of steps at a word $w_{\text{red}} \in \mathcal{R}(X)$ with $[w] = [w_{\text{red}}]$.

Injectivity (Uniqueness via van der Waerden's Action): To show that distinct reduced words cannot be congruent under $\widehat{R}$, let $\text{Sym}(\mathcal{R}(X))$ be the symmetric group of all permutations of the set $\mathcal{R}(X)$.

For each letter $x \in X$, define a map $\sigma_x : \mathcal{R}(X) \to \mathcal{R}(X)$ by:

$$\sigma_x(u_1 u_2 \cdots u_n) = \begin{cases} x u_1 u_2 \cdots u_n & \text{if } u_1 \neq x', \\ u_2 \cdots u_n & \text{if } u_1 = x'. \end{cases}$$

Similarly, define $\sigma_{x'} : \mathcal{R}(X) \to \mathcal{R}(X)$ by:

$$\sigma_{x'}(u_1 u_2 \cdots u_n) = \begin{cases} x' u_1 u_2 \cdots u_n & \text{if } u_1 \neq x, \\ u_2 \cdots u_n & \text{if } u_1 = x. \end{cases}$$

Direct computation shows that $\sigma_x \circ \sigma_{x'} = \text{id}_{\mathcal{R}(X)}$ and $\sigma_{x'} \circ \sigma_x = \text{id}_{\mathcal{R}(X)}$. Therefore, $\sigma_y \in \text{Sym}(\mathcal{R}(X))$ for all $y \in \overline{X}$.

By the universal property of the free monoid $\overline{X}^*$, the assignment $y \mapsto \sigma_y$ extends to a unique monoid homomorphism $\Theta : \overline{X}^* \to \text{Sym}(\mathcal{R}(X))$. For each $x \in X$:

$$\Theta(xx') = \sigma_x \circ \sigma_{x'} = \text{id} = \Theta(1) \quad \text{and} \quad \Theta(x'x) = \sigma_{x'} \circ \sigma_x = \text{id} = \Theta(1).$$

Thus, if $u \ \widehat{R} \ v$, then $\Theta(u) = \Theta(v)$ in $\text{Sym}(\mathcal{R}(X))$.

Now suppose $u = u_1 \cdots u_n$ and $v = v_1 \cdots v_m$ are two reduced words such that $[u] = [v]$ in $FG(X)$. Then $\Theta(u) = \Theta(v)$. Applying both permutations to the empty word $1 \in \mathcal{R}(X)$:

$$\Theta(u)(1) = \sigma_{u_1} \cdots \sigma_{u_n}(1) = u_1 u_2 \cdots u_n = u,$$

$$\Theta(v)(1) = \sigma_{v_1} \cdots \sigma_{v_m}(1) = v_1 v_2 \cdots v_m = v.$$

Thus, every element of $FG(X)$ is represented by a unique reduced word. □

1.7.3 Counting Reduced Words of Length n

Let $|X| = k \geq 1$. The extended alphabet contains $|\overline{X}| = 2k$ symbols.

- **Length 0:** There is 1 reduced word: the empty word 1.
- **Length 1:** Any of the $2k$ symbols can be chosen: $2k$ reduced words.
- **Length $n \geq 2$:** A reduced word $u_1 u_2 \cdots u_n$ must satisfy $u_{j+1} \neq u_j'$. Thus, the first letter u_1 has $2k$ choices, and each subsequent letter u_{j+1} can be any symbol in $\overline{X}$ *except* the formal inverse of the immediately preceding symbol u_j' (leaving $2k - 1$ choices).

By the rule of product, the number of distinct reduced words of length $n \geq 1$ is:

$$N(n) = 2k(2k-1)^{n-1}.$$

Because every reduced word represents a unique element of $FG(X)$, all these words correspond to **pairwise distinct** elements in the group.

1.8 Generalized Direct Sum $\mathbb{Z}^{\oplus X}$

Definition 1.11 (Free Abelian Group $\mathbb{Z}^{\oplus X}$)**.** Let X be an arbitrary set. The **free abelian group on** X, denoted by $\mathbb{Z}^{\oplus X}$ (or $\mathbb{Z}^{(X)}$), is defined as the set of all functions $\alpha : X \to \mathbb{Z}$ with **finite support**:

$$\mathbb{Z}^{\oplus X} := \{\alpha : X \to \mathbb{Z} \mid \operatorname{supp}(\alpha) = \{x \in X \mid \alpha(x) \neq 0\} \text{ is finite}\}.$$

Equipped with pointwise addition $(\alpha + \beta)(x) = \alpha(x) + \beta(x)$, $\mathbb{Z}^{\oplus X}$ forms an abelian group whose identity is the zero function.

For each $x \in X$, let $e_x \in \mathbb{Z}^{\oplus X}$ denote the indicator function:

$$e_x(y) = \begin{cases} 1 & \text{if } y = x, \\ 0 & \text{if } y \neq x. \end{cases}$$

Every $\alpha \in \mathbb{Z}^{\oplus X}$ can be uniquely written as a finite linear combination:

$$\alpha = \sum_{x \in \operatorname{supp}(\alpha)} \alpha(x) e_x.$$

Proposition 1.5 (Relation between $FG(X)$ and $\mathbb{Z}^{\oplus X}$)**.** *Let $[FG(X), FG(X)]$ denote the derived (commutator) subgroup of $FG(X)$. Then the abelianization of the free group is isomorphic to the free abelian group:*

$$FG(X)/[FG(X), FG(X)] \;\cong\; \mathbb{Z}^{\oplus X}.$$

Proof. Define a set map $j : X \to \mathbb{Z}^{\oplus X}$ by $j(x) = e_x$. By the universal property of $FG(X)$, there exists a unique homomorphism $\pi_{\text{ab}} : FG(X) \to \mathbb{Z}^{\oplus X}$ satisfying $\pi_{\text{ab}}([x]) = e_x$. Because $\mathbb{Z}^{\oplus X}$ is abelian, $[FG(X), FG(X)] \subseteq \ker(\pi_{\text{ab}})$.

Thus π_{ab} factors through the quotient group $FG(X)/[FG(X), FG(X)]$ to yield a surjective homomorphism $\bar{\pi} : FG(X)/[FG(X), FG(X)] \to \mathbb{Z}^{\oplus X}$. The inverse map is constructed by sending the basis $\{e_x\}$ back to the coset $[x][FG(X), FG(X)]$, establishing the isomorphism. □

Since for any map f from X to an abelian group G, the kernel of the induced map contains $[FG(X), FG(X)]$, hence it factors to give a unique homomorphism to $\mathbb{Z}^{\oplus X}$. This gives us the universal property for $\mathbb{Z}^{\oplus X}$ on abelian groups.

1.9 Presentations of Groups

Definition 1.12 (Group Presentation)**.** Let X be a set and let $R \subseteq FG(X)$ be a set of elements (called **relators**).

(a) The **normal closure** of R in $FG(X)$, denoted by $\langle\langle R \rangle\rangle$, is the smallest normal subgroup of $FG(X)$ containing R:

$$\langle\langle R \rangle\rangle := \bigcap_{\substack{N \trianglelefteq FG(X) \\ R \subseteq N}} N = \left\{ \prod_{i=1}^{m} g_i r_i^{\pm 1} g_i^{-1} \;\middle|\; m \geq 0,\, g_i \in FG(X),\, r_i \in R \right\}.$$

(b) The group **presented by generators X and relators R**, denoted by $\langle X \mid R \rangle$, is the quotient group:

$$\langle X \mid R \rangle := FG(X)/\langle\langle R \rangle\rangle.$$

When a relation is expressed as $u = v$ (or $u \sim v$), it corresponds to the relator $v^{-1}u \in R$.

Theorem 1.6 (Universal Property of Group Presentations)**.** *Let $G = \langle X \mid R \rangle$, and let $\pi_R : FG(X) \to G$ be the quotient homomorphism. Let H be an arbitrary group and let $f : X \to H$ be a set map. Let $\widetilde{f} : FG(X) \to H$ be the unique homomorphic extension of f.*

*If $\widetilde{f}(r) = 1_H$ for all $r \in R$, then there exists a **unique** group homomorphism $\phi : G \to H$ such that $\phi([x]) = f(x)$ for all $x \in X$.*

Proof. Follows from Theorem 1.3 and Exercise 1.8. □

1.10 Examples of Presentations

A standard procedure applies to establish that a given presentation $\langle X \mid R \rangle$ is isomorphic to a concrete finite group G:

1. Construct a homomorphism $\Phi : \langle X \mid R \rangle \to G$ using the Universal Property of Presentations, and show Φ is surjective, giving $|\langle X \mid R \rangle| \geq |G|$.
2. Use the relations R to systematically reduce any formal word into a bounded list of normal forms, establishing an upper bound $|\langle X \mid R \rangle| \leq |G|$.
3. Conclude that Φ is a bijection, and hence an isomorphism.

1.10.1 Presentation of S_2

Proposition 1.6. *The symmetric group on two letters $S_2 \cong \mathbb{Z}_2$ has the presentation:*

$$S_2 \;\cong\; \langle s \mid s^2 = 1 \rangle.$$

Proof. Let $G = \langle s \mid s^2 = 1 \rangle = FG(\{s\})/\langle\langle s^2 \rangle\rangle$.

1. **Surjective Homomorphism onto S_2:** In $S_2 = \{(1), (1\,2)\}$, the transposition $\sigma = (1\,2)$ satisfies $\sigma^2 = \mathrm{id}$. Define $f : \{s\} \to S_2$ by $f(s) = \sigma$. Its extension satisfies $\widetilde{f}(s^2) = \sigma^2 = \mathrm{id}$. By the universal property of presentations, this induces a well-defined homomorphism $\Phi : G \to S_2$ with $\Phi([s]) = \sigma$. Since σ generates S_2, Φ is surjective, so $|G| \geq |S_2| = 2$.
2. **Upper Bound on Order:** In G, the relation $s^2 = 1$ implies $s^{-1} = s$. Any word in $FG(\{s\})$ is of the form s^n ($n \in \mathbb{Z}$). Reducing modulo $s^2 = 1$ shows $[s^n] = [s^{n \bmod 2}]$. Thus, every element of G lies in the set $\{[1], [s]\}$. Hence $|G| \leq 2$.

□

1.10.2 Presentation of S_3

Proposition 1.7. *The symmetric group on three letters S_3 has the presentation:*

$$S_3 \;\cong\; \langle s_1, s_2 \mid s_1^2 = 1,\; s_2^2 = 1,\; (s_1 s_2)^3 = 1 \rangle.$$

Proof. Let $G = \langle s_1, s_2 \mid s_1^2 = 1,\, s_2^2 = 1,\, (s_1 s_2)^3 = 1 \rangle$.

1. **Surjective Homomorphism onto S_3:** In S_3, consider the adjacent transpositions $\sigma_1 = (1\,2)$ and $\sigma_2 = (2\,3)$. We compute directly:

   $$\sigma_1^2 = \mathrm{id}, \quad \sigma_2^2 = \mathrm{id}, \quad \sigma_1\sigma_2 = (1\,2\,3), \quad (\sigma_1\sigma_2)^3 = (1\,2\,3)^3 = \mathrm{id}.$$

   Define $f : \{s_1, s_2\} \to S_3$ by $f(s_1) = \sigma_1$ and $f(s_2) = \sigma_2$. All three relators evaluate to id in S_3, so by the universal property of presentations, there exists a unique homomorphism $\Phi : G \to S_3$ sending $[s_1] \mapsto \sigma_1$ and $[s_2] \mapsto \sigma_2$. Because adjacent transpositions generate S_3, Φ is surjective, which implies $|G| \geq |S_3| = 6$.

2. **Upper Bound on Order:** The relations $s_1^2 = 1$ and $s_2^2 = 1$ imply $s_1^{-1} = s_1$ and $s_2^{-1} = s_2$. Thus, any element of G can be written as an alternating product of s_1 and s_2. The relation $(s_1 s_2)^3 = 1$ expands to $s_1 s_2 s_1 s_2 s_1 s_2 = 1$, which is equivalent to the braid relation:

$$s_1 s_2 s_1 = s_2 s_1 s_2.$$

This relation implies that any alternating subword of length 3 can be flipped. Consequently, any word of length ≥ 4 can be reduced:

$$s_1 s_2 s_1 s_2 = (s_1 s_2 s_1) s_2 = (s_2 s_1 s_2) s_2 = s_2 s_1 (s_2^2) = s_2 s_1,$$

$$s_2 s_1 s_2 s_1 = (s_2 s_1 s_2) s_1 = (s_1 s_2 s_1) s_1 = s_1 s_2 (s_1^2) = s_1 s_2.$$

Thus, every element of G is represented by an alternating word of length at most 3:

$$G = \{[1],\ [s_1],\ [s_2],\ [s_1 s_2],\ [s_2 s_1],\ [s_1 s_2 s_1]\}.$$

Therefore, $|G| \leq 6$.

Since $|G| \geq 6$ and $|G| \leq 6$, $|G| = 6$. Being a surjective map between groups of order 6, Φ is an isomorphism, so $G \cong S_3$. □

1.10.3 Presentation of the Dihedral Group D_{2n}

Definition 1.13 (Dihedral Group D_{2n})**.** The dihedral group D_{2n} is the group of symmetries of a regular n-gon, containing $2n$ elements: n rotations and n reflections. For $n = 3$, D_6 is the symmetry group of an equilateral triangle (which is isomorphic to S_3).

Proposition 1.8. *The dihedral group of order $2n$, D_{2n}, has the presentation:*

$$D_{2n} \ \cong\ \langle a, b \mid a^n = 1,\ b^2 = 1,\ bab = a^{-1} \rangle.$$

Proof. Let $G = \langle a, b \mid a^n = 1,\ b^2 = 1,\ bab = a^{-1} \rangle$.

1. **Surjective Homomorphism onto D_{2n}:** Define $f(a) = r$ (rotation by $\frac{2\pi}{n}$) and $f(b) = s$ (reflection); clearly f induces a homomorphism from G to D_{2n} since $r^n = 1$, $s^2 = 1$, and $srsr = 1$. Also, since r, s are generators of D_{2n}, it is surjective.

2. **Upper Bound on Order:** The relation $sr = r^{-1}s$ implies $sr^k = r^{-k}s$. Using this identity, any occurrence of s followed by r can be rewritten with powers of r on the left and s on the right. Thus, every element $[w] \in G$ can be simplified to the normal form:

$$[r^i s^j] \quad \text{with } i \in \{0, 1, \ldots, n-1\} \text{ and } j \in \{0, 1\}.$$

The number of distinct elements is bounded by $n \times 2 = 2n$. Hence $|G| \leq 2n$.

Combining both bounds yields our result. □

1.10.4 Presentation of the Quaternion Group Q_8

Definition 1.14 (Quaternion Group Q_8)**.** The quaternion group Q_8 is the order 8 non-abelian group given by the set of unit quaternions:

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

satisfying $i^2 = j^2 = k^2 = ijk = -1$, with -1 commuting with all elements.

Proposition 1.9. *The quaternion group Q_8 has the presentation:*

$$Q_8 \cong \langle x, y \mid x^4 = 1,\ x^2 = y^2,\ yxy^{-1} = x^{-1} \rangle.$$

Proof. Let $G = \langle x, y \mid x^4 = 1,\ x^2 = y^2,\ yxy^{-1} = x^{-1} \rangle$.

1. **Surjective Homomorphism onto Q_8:** In Q_8, identify $x \mapsto i$ and $y \mapsto j$. We verify the relations:

   $$i^4 = (-1)^2 = 1, \quad i^2 = -1 = j^2, \quad jij^{-1} = ji(-j) = j(k) = -i = i^{-1}.$$

   All three relators are satisfied in Q_8. By the universal property of presentations, there exists a unique homomorphism $\Phi : G \to Q_8$ sending $[x] \mapsto i$ and $[y] \mapsto j$. Since i and j generate Q_8, Φ is surjective, so $|G| \geq |Q_8| = 8$.

2. **Upper Bound on Order:** From $yxy^{-1} = x^{-1}$, multiplying by y on the right gives:

   $$yx = x^{-1}y = x^3y.$$

   This rewriting rule pushes y to the right of x. Moreover, $y^2 = x^2$ implies that any even power of y converts into an even power of x:

   $$y^2 = x^2, \quad y^3 = x^2y, \quad y^4 = x^4 = 1.$$

   Consequently, any word in x and y can be expressed in the form:

   $$[x^a y^b] \quad \text{where } a \in \{0, 1, 2, 3\} \text{ and } b \in \{0, 1\}.$$

   There are at most $4 \times 2 = 8$ such cosets:

   $$G = \{[1], [x], [x^2], [x^3], [y], [xy], [x^2y], [x^3y]\}.$$

   Hence $|G| \leq 8$.

□

Exercise 1.15. Every group G occurs as a quotient of a free group over some set X. If G is finite, then X can be taken to be finite.
Hint: Take $X = G$ and R to be the group table of G.

Theorem 1.7. *If H is a subgroup of the free group $FG(X)$, then H is also a free subgroup.*

Proof to be completed. □

References

1. Dummit, D. S., and Foote, R. M., *Abstract Algebra*, 3rd ed., John Wiley & Sons, 2004.

2. Lang, S., *Algebra*, 3rd ed., Graduate Texts in Mathematics, Springer-Verlag, 2002.

Further references will be added as the notes develop.